Designing and Evaluating Adaptive Controllers for People with Limited Mobility

An Interactive Qualifying Project
submitted to the Faculty of

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Abstract

Individuals with limited motor mobility tend to encounter physical barriers with everyday activities. Foremost among these is the process of musical creation, in which lack of physical movement hinders the capacity to play standard instruments effectively. This project intended to design two adaptive controllers, each capable of fostering more accessible musical fulfillment and a broader sense of personal autonomy for individuals with limited mobility. Once prototyped, an interview was conducted with quadriplegic musician and inventor Dave Whalen to evaluate general community applicability for each design.

Executive Summary

For individuals living with high-level motor impairments such as quadriplegia, standard instruments tend to be difficult to use. Luckily, accessible musical instruments sustain creative expression for many niches within this community. Throughout the research process for this project, it became clear that this abundance of technologies was prompted by an impassioned community of musicians, scholars, and inventors. While it might be possible to improve upon an existing instrument, few controllers exist that implement a hybrid design capable of addressing needs beyond just music. Deriving inspiration from both accessible musical instruments and standard adaptive controllers, I designed two primitive hybrid controllers capable of musical use and common task commands. The first design, “Prototype I,” incorporates sip & puff mechanics using breath inputs to allow 6 outputs. The second design, “Prototype II,” offers a vocal alternative, utilizing a microphone to trigger computer commands or musical notes. To assess the viability of community adaptation, an evaluative interview was conducted with quadriplegic musician and inventor, Dave Whalen. His insight elucidated the need for a wider range of outputs and musical expression, likely indicating a capacity over three times larger than current designs

permit. Mr. Whalen also identified other challenges within the designs, such as difficulty individuals would face when customizing their own setup. Still, he was optimistic about the designs and believes there exists a place for community adoption. Together with Mr. Whalen's inputs, I compiled recommendations for potential future studies and design innovations. Most notably, the range of outputs could be expanded simply by adding a secondary input port to either prototype design. Moreover, the less-intuitive aspects of setup could be avoided by incorporating a graphical user interface into the prototype itself, even if not addressing any blatant physical barriers. Overall, the designs are imperfect but remain conducive for personal independence, with future iterations likely granting a larger range of expression while improving accessibility.

Introduction

Traditional musical instruments often create physical barriers for individuals with mobility issues. For someone with severe physical impairments, these challenges may deprive them of creative musical expression and cause difficulty with everyday activities. Due to this, research and commercial groups frequently develop accessible instruments, ranging from the Magic Flute to MotionComposer. This is true in other areas as well, with adaptive technology in genres such as gaming and home control units. However, these devices often incur a range of operational and commercial success, with output latency and motor fatigue limiting the potential for many accessible musical instruments. This project intends to deconstruct some of these limitations by proposing two adaptive controller designs for common activities, with the option for musical use. Of the many sources that have provided insight for this project, notable influences include the Jamboxx for its approach to breath amplitude, Origin Instrument's "Breeze" product, which implements sip-and-puff mechanics, and Holland Bloorview's "The Hummingbird", which converts simple speech inputs into usable commands. However, assessing real-world validity of the controllers expanded beyond

relying on previous technology; a qualitative interview was conducted with quadriplegic musician and adaptive controller enthusiast Dave Whalen. Designed to analyze community viability, Mr. Whalen's insight helped inform a more holistic understanding of adaptive controllers.

Background

This section will examine the basic groundwork of adaptive controllers in the limited mobility community. It will include a description of the accessibility barriers for individuals with motor impairment when playing standard musical instruments, along with a background on physical considerations for someone within this community. This will be followed by context on previous technological innovations.

Accessibility Barriers & Physical Considerations

Instruments often require relatively challenging motor movements, becoming impossible to play for individuals with severe motor disabilities. Typical acoustic instruments are characterized by things like the use of multiple limbs, motor dexterity, advanced muscular movements, and physical endurance for sustained playing (Dixon, 2003). With the piano, for instance, a user must coordinate their fingers to press down weighted notes. For someone with a severe physical disability, playing even a simple piece can be formidable. In this case, difficulties can stem from an inability to press down on keys or poor motor skills, along with numerous other physical barriers. Many are therefore systemically excluded from any involvement in the music-making process, even if inadvertently.

Beyond physical considerations, music is often valued for creative expression and self-fulfillment. Published in the journal of the Canadian Disability Studies Association, the research frames music engagement as an intrinsic human experience (Grond et al., 2020). However, the loss of playing an instrument can cause someone to lose an important means of creative

expression and, more importantly, self-agency. This is not amended by enforcing an individual to undergo severe physical rehabilitation, which is as inequitable as it is impractical. Instead, numerous studies assert that fulfillment should come through reinforcing the music-making process. This idea is reflected in the design of assistive musical instrument interfaces, which adapt to a user's specific physical capabilities (Frid, 2019).

Based on these criteria, creating an assistive musical instrument can be difficult, particularly one that is effective. Much of this interdisciplinary research and prototyping therefore tends to be institution driven. For instance, the RATlab – a research collective at Worcester Polytechnic Institute (WPI) under the direction of Professor V.J. Manzo – conjoins knowledge from various academic and occupational backgrounds to create non-traditional and adaptive musical systems (Manzo, n.d.). Similarly, a cross-disciplinary team at McGill University has compiled an exploration of twenty years of narrative and intervention studies on adaptive musical instruments (Grond et al., 2020). Others from within the limited-mobility community provide experienced feedback and greatly assist with prototype development. Fittingly, the inspiration for many adaptive musical instruments comes directly from people who see applications for those in similar situations as themselves.

Regardless of origin, a user should be identified for any adaptive controller to understand what specific physical accommodations are needed. For most accessible instruments and adaptive controllers, the broad community includes people with quadriplegia or other conditions that limit fine motor control. It should be noted, however, that, even within this demographic, individuals exhibit varying degrees of movement capacity and control. This makes assistive devices intrinsically incapable of catering to every need, instead opting to focus on whatever specific, non-standard motor patterns an individual can make without physical strain (Wobbrock et al., 2018). As such, developers are naturally required to adhere to personalized evaluation criteria, even if not with explicit intent.

Prior Technological Innovations

Many niches exist within the umbrella of accessible instruments, where controllers have been designed to address virtually every physical need. Notable among these are touch-based controllers, which have relatively low physical barriers. Devices such as the Skoog, a soft and squeezable foam cube with color-coded buttons, allow users with physical or learning disabilities to trigger notes by simple touch movements (Campbell, 2020). Similarly, the SenseEgg utilizes a handheld, egg-shaped controller with internal motion sensors and rangefinders to translate hand motions into synthesized sounds, like that of a theremin (Blatherwick et al., n.d.). The TouchTone, a pressure-sensitive alternate controller created for children with conditions like cerebral palsy, involves one hand pressing a highly sensitive pad to trigger a note while the other shifts octaves (Bhat, 2010). Notably, however, these devices cannot be used by individuals who experience severe difficulty moving their limbs.

For people in this situation, motion-capturing controllers can instead translate limited body movements into sound. For instance, the Soundbeam uses ultrasonic sensors that release an invisible beam of light, triggering a sound when something intersects with the beam. Another controller, the MotionComposer uses 3D camera sensors that track body movement for sound. Even free, open-source software is available, including the Adaptive Use Musical Instrument, utilizing a comparable camera system. (Ward et al., 2019) While these devices can be used without much physical movement, they commonly face another problem: latency. Cameras take time to convert video into a corresponding sound, creating a noticeable lag between movement and output, called latency. In the context of music, high latency can disrupt a user's timing and engagement. Further, motion-capturing devices can be less useful for those with more restricted movements.

For those with more significant mobility impairments, eye-tracking, breath, and voice alternatives also exist. In the mid-2000s, the Magic Flute was first tested as an electronic wind controller designed for individuals with limited hand movements, but who could retain head movement and breath

control. Users can play music by blowing into a tube for volume (where blowing harder produces a larger sound output) and tilt their head up and down to determine pitch (The Magic Flute II, n.d.). Comparatively, the Flote is a wind instrument that uses breath pressure to dictate note volume and intensity. Like the Magic Flute, it uses head tilts to determine individual notes (Aziz et al., 2008). Varying slightly, Netytar is a software-based design that combines gaze tracking (for note selection) with a breath sensor for note pitch, duration, and dynamics. Commercial popularity improved in the space with the development of the "Jamboxx," an electronic harmonica-style MIDI controller. Co-created by quadriplegic musician Dave Whalen, the device uses breath amplitude for sound and mouthpiece movement for note pitch (Jamboxx, n.d.). Still, many existing accessible musical instruments remain expensive and unable to be used beyond musical purposes.

Even so, other forms of adaptive controllers have been developed. For all intents and purposes, this paper will focus on breath-controlled examples. Basic multi-command sip-and-puff switches, such as the "Breeze" by Origin Instruments, offer notable alternatives to simple "on/off" buttons. Here, advanced pneumatic switches measure both direction (in/out) and amplitude (hard/soft) of breath (Sip/puff Switch, n.d.). Typically, this gives the user four distinct outputs from a single tube: strong exhale, gentle exhale, strong inhale, and gentle inhale. A "sip" is considered an inhale while a "puff" is an exhale. Sip-and-puff mechanics exist in numerous breath switches and will be common in many subsequent controllers. Environmental control units provide another interesting use of breath inputs, in which a screen or hub can cycle through pre-set options (for instance, lamp, TV, thermostat, front door), issuing a specific breath pulse when the desired option is ready (Assistive Technology Glossary, 2017). Next, adaptive gaming controllers offer some notable inclusivity alternatives to traditional console and PC controls. The Xbox Adaptive Controller, for instance, is compatible with sip-and-puff switches (Using Switches to Access Button Presses, n.d.), while the QuadStick – designed specifically for quadriplegic gamers – allows a user to map an entire Xbox or PlayStation controller to their mouth (Quadstick User

Guide, n.d.). For people who have limited breath pressure but can still make sounds, acoustic amplitude switches are an appropriate alternative. Rather than measuring air pressure in a tube, these devices measure the amplitude (volume) of a vocalization through a microphone. One such switch is Holland Bloorview's "The Hummingbird" (Holland Bloorview's Hummingbird, 2022).

Evidently, an abundant number of accessible musical instruments and other controllers exist for various niches within the limited-mobility community. Many of these very effectively address the demands of their existence. As such, creating another device for one narrow application would not necessarily be useful. Rather than trying to improve existing technology or invent a new niche entirely, this project focuses on potential applications for a general individual in the community. Specifically, it focuses on designing a prototype ergonomic hybrid controller capable of carrying out pre-programmed commands while allowing a user to play a novice form of music. Ideally, this type of controller would allow its users to experience a form of greater self-autonomy with their actions.

Methodology

This section will outline the strategic framework used throughout the course of the project, sectioned into a basic evaluative criteria component followed by two main phases. The first involves a multifaceted development process for two potential controller prototypes, intended for individuals with limited mobility to carry out everyday tasks to allow music engagement. This portion will encompass the research and design components of the project. The second section recounts a semi-structured interview with Dave Whalen, an individual with extensive experience developing and using adaptive controllers. This segment assists in qualitatively evaluating the general applicability of each prototype.

Evaluation Criteria

To understand the criteria necessary for each design, I first created a series of personal general evaluation questions to consider. While being unable to accurately address them all without a tangible physical prototype, they helped to guide the focus of each design, illustrated below:

I. What features are especially comfortable or intuitive to use?
II. What is the range of expression permitted by the prototype (digital commands, dynamics, notes)?
III. How much exertion is required to achieve tangible outputs?
IV. How effectively does the design precipitate movements the user can reliably control?
V. What are the practical trade-offs between a simple and intuitive design focus compared to a wider range of use and expression?
VI. Are there any potential sharp edges, pressure points, or anything that would otherwise risk strain/overexertion?
VII. What challenges might a user encounter when trying to set up and use the device independently?

These questions were created prior to either prototype design, so should not contain bias. Moreover, although they were made before any substantial design progress, their criteria were still useful for approaching the interview and for personal evaluation. As such, I chose to also review them in hindsight once completed; this section was included along with the results later in the paper. Early on, however, the evaluative questions existed to provide a foundation to proceed from.

The Design Process

The general idea for Prototype I was to design an ergonomic system that could translate breath pressure into standard computer commands (like keyboard shortcuts), which could then be mapped to do almost anything using macro software. First, the controller would require a sensor capable of

detecting inhale and exhale intensity. With simple wiring, it was determined that differential pressure sensors such as the MPX5010DP and MPS20N0040D could be configured to microcontrollers such as Arduino Leonardo or Raspberry Pi Pico to transport data. These microcontrollers would essentially act as an intermediary device between the pressure sensors and a laptop. Alternatively, buying a direct sip-and-puff switch such as Origin Instruments "Breeze" could work as well, already translating breath into standard computer commands. Assuming cost efficiency, the microcontroller setup would be more optimal; in which case, the next step would be to code the microcontroller to define pressure trigger zones. Instead of using the basic four sip-and-puff triggers, this controller could utilize six for greater versatility. These would include a soft puff, hard puff, soft sip, hard sip, double puff, and double sip. Once plugged into the computer via USB, additional code would then divide each of these into representative keyboard keys, such as F1 through F6 due to common lack of use. Automation programs on the computer like AutoHotkey or Keyboard Maestro could be set up to initiate physical commands when keyboard triggers are invoked. These might range anywhere from typing an email template to playing a chord on the piano. Notably, there would be only six possible outputs in this design, so configuring the commands into a standard musical scale – which has seven distinct notes – is not possible. I generated a diagram below of what the design might look like during use.

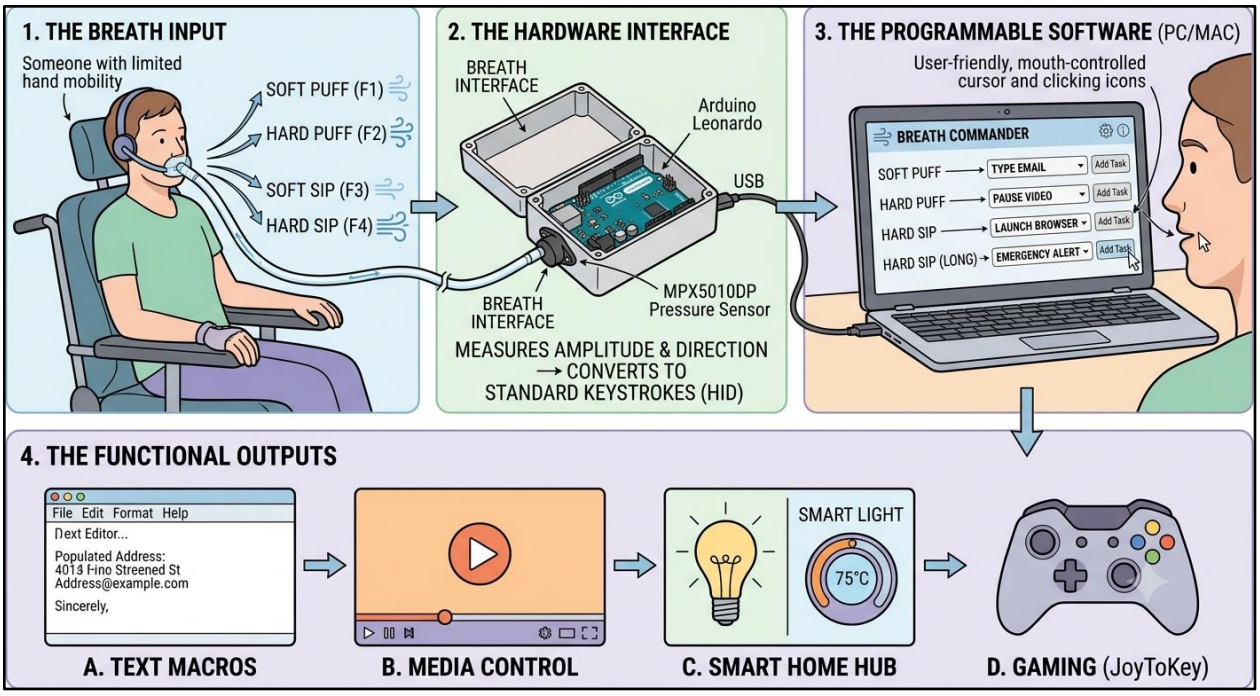


Image generated by Google Gemini

Prototype II would utilize a similar setup as the first prototype but instead focus on an ergonomic microphone as the input device. In this model, a standard microphone headset would be plugged into a sound amplifier module, such as a MAX4466 or MAX9814. These act as small circuit boards with built-in microphone jacks, which would boost the audio into a usable electrical voltage. Simple wiring could then be used to plug into either of the microcontrollers previously mentioned, the Arduino Leonardo or Raspberry Pi Pico. While the context would be different from the previous controller, the functionality from this point onward remains nearly identical to Prototype I. One end of a standard USB cable would be plugged into the port of the microcontroller while the other end would be plugged into the USB port on a computer. Instead of sip-and-puff, the microcontrollers would now be coded to measure types of sounds. Thresholds might include a short and soft hum, a sharp tongue click, and any sustained sound. In an effort not to detect background noise, frequency filters could be placed into the code as well. Similar to the previous design, these triggers would be coded to denote certain lesser-used keyboard keys, such as F1 through F3. Computer programs like AutoHotkey or Keyboard Maestro could create commands for

these keys, with both common and musical uses. The drawback here would be that, due to fewer possible triggers, scope of use becomes even more limited than with the previous design. An illustration of what this might look like is provided below.

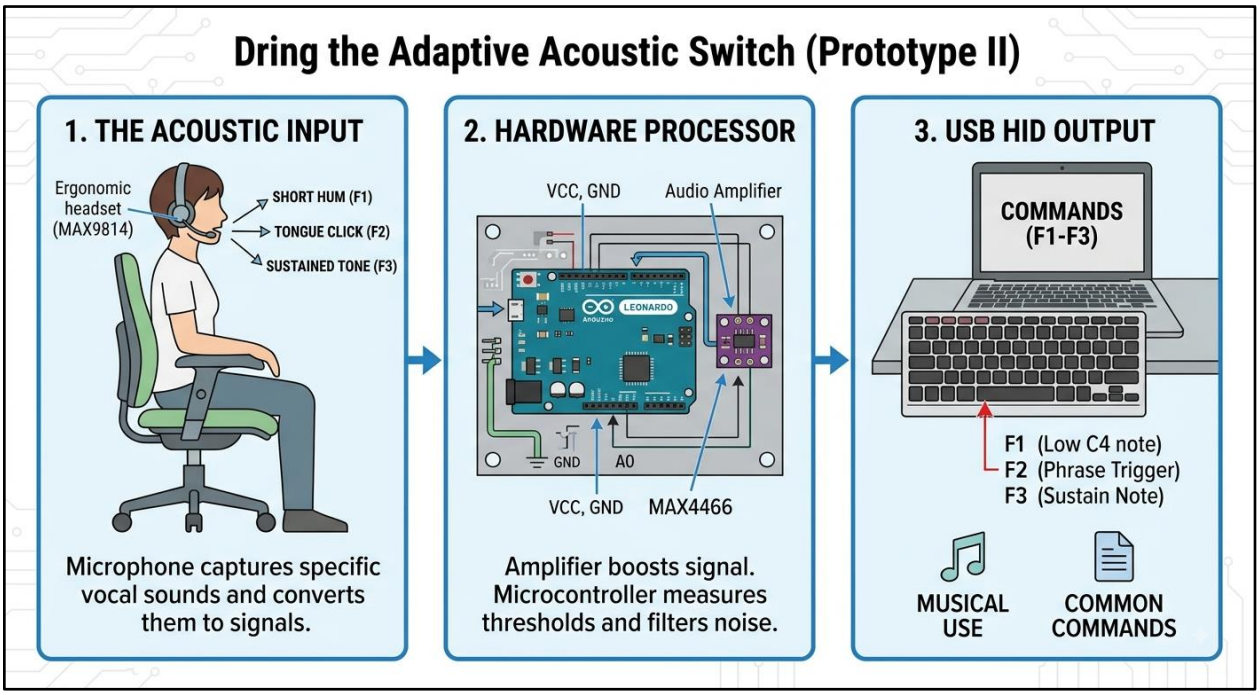


Image generated by Google Gemini

The Interview

To partially evaluate how the community might respond to these prototypes, I conducted an interview with Dave Whalen on a video call. Engrained in the development of accessible musical instruments like Jamboxx, Mr. Whalen is also an experienced quadriplegic musician with a reputable perspective on adaptive controllers. During the interview, the prototypes were explicitly stated as “still in progress”. By emphasizing the flexibility of each design, I wanted to focus on constructive optimizations.

The questions themselves focused on general applicability, mostly avoiding personal preference or physical limitations. For instance, I mentioned that Prototype II utilizes sound duration and volume for data before asking, “What unique advantages or disadvantages do you foresee when using vocal thresholds for individuals within the limited-mobility

community?" This question frames the applicability of the microphone setup within a community context. The interview also explicitly included open-ended questions rather than simple "yes" or "no" questions to encourage answers with more depth. For example, instead of simply asking if a breath interface would be useful for the general community, this was instead framed as "how useful" it might be. Responses that would have otherwise been rigid became more nuanced, including ones that do not strongly conform to either end of answer extremes.

Beyond the actual questions, the interview had a simple framework. It opened with a brief background section on the intentions of the interview and what Mr. Whalen's role would be. Specifically, this involved identifying how the prototypes might serve a general audience and distinguishing possible ergonomic adjustments. I then introduced Prototype I, explaining what it is, how data gets mapped, and what it controls. When complete, three basic questions were asked about the design, with additional follow-up questions according to the answers provided. This process was repeated for Prototype II as well, with altered questions to better address the design on its own merits. Once complete, a separate questioning section took place to compare the prototypes. For instance, Mr. Whalen was asked to compare exertion on the breath-pressure prototype to that of the vocal threshold design for the general community. This question directly compares the two designs, without invoking too much personal preference.

Data & Analysis

This section presents the qualitative results and findings from the semi-structured interview with quadriplegic musician and innovator, Dave Whalen. The interview focused on using his background expertise and life experiences to evaluate each prototype design. Particularly, it involved analyzing components of each design separately, followed by collective feedback and direct comparison.

Prototype I

One of the main objectives of this stage of the interview was to determine if a breath-pressure system would be a feasible adoption for individuals with limited motor use. Regarding this, Dave asserted he “was thinking about this spinal injured population that are quadriplegic... it's one of the kind of few technologies that you could use.” He went on that physical switches often require active head motion or tapping, precipitating movements which some individuals are incapable of. Mr. Whalen also introduced that a large threshold of notes could be programmed into, for instance, a single breath. Hypothetically, a small exhale could trigger a low note while a large exhale could trigger a high note. With this approach, Dave believed the breath interface could be accompanied by some variant of visual feedback to inform the user of note qualities. “You could see when you're striking a note... You could practice maybe to learn the velocity... You could certainly do a hard and soft. That would not be hard. You could probably do 3 [octaves] at least." Lastly for this design, when asked whether users in the quadriplegic community would be able to perform independent setup, the response was a resounding “No.” He believed it important that devices maintain a baseline, possibly with both a physical configuration and with default software presets.

Prototype II

Mr. Whalen characterized the microphone setup as a possibly cheaper alternative in contrast with the first design. Still, he accentuated that standard microphones and speech-recognition tend to fail in live or noisy settings. When asked about bypassing background noise, Dave suggested isolating vocal inputs by placing the microphone inside the mouth. “You could use your own vocal cords as part of the instrument... if I was humming in a crowd, no one would hear me... but if you could amplify and isolate the sound in a new way, you could get the musical part out of the person.” He compared this to internal vibrations on instruments, such as placing a microphone on a harmonica or using an electric kazoo. As a more extensive

option than using the voice to trigger digital commands, Dave offered that the mouth could act as an acoustic filter for external sounds. He highlighted how Peter Frampton facilitates a guitar sound into a tube which goes into his mouth, then using his mouth to shape the sound into a microphone. Contextualizing this for an adaptive design might involve using a tube to transport synthesized sounds or presets into a user's mouth, where mouth movements can morph the sound into a microphone. To trigger notes, Dave suggested conjoining the vocal setup with external physical switches. “You could trigger phrases, and then shape them with your mouth... A lot of disabled people could use the keyboard, use the white keys as adaptive switches. They're not required to sit there with their fingers... But they could use like a pencil or their hand... and just push a key."

Collective Feedback

During the interview, Dave emphasized the need for a larger range of music outputs. He cited that, playing pieces such as “Somewhere Over the Rainbow” or the National Anthem (which he has performed on the Jamboxx for an Albany Empire football game), a user would need at least 2.5 to 3 octaves. Essentially, this means that instead of the 3-6 outputs possible in the current designs, there would need to be closer to 20. I asked Dave if he had any thoughts on how this could be achieved, given the schematics of the prototypes. We discussed a few options, including incorporating a second pressure sensor or microphone, which would essentially double the possible outputs. Another idea was to combine the already existing outputs with motion tracking, such as head or nose. Referring to Prototype I, Dave said that “Trackers are slow... they can be slow, as long as the breath is fast, you're okay.” Finally, he offered that perhaps chords and strumming patterns could be presets, where single breath inputs would trigger more complex outputs. This tends to be a double-edged scenario, where more variety of expression may be possible, but the user does not control as much of what comprises that expression. Dave highlighted that this might be an instance where future innovators may find a better solution.

Comparison

Towards the conclusion of the interview, Mr. Whalen answered a few brief questions comparing the two designs. When asked about physical exertion with the prototypes, he stated that anything vocal is much more difficult. This did, however, include an innuendo in that using breath flow instead of simple breath pressure would induce much higher exertion levels. In terms of adoptability, Dave favored the vocal setup this time, citing the expansive use of microphones and their compatibility. Finally, he seemed optimistic that the designs could address both different and similar needs in audiences. “They might be able to deliver unique things. They also might be able to be combined.” Dave seemed to believe that choosing one design over the other is ultimately unnecessary, and that the prototype used might simply depend on personal preference or specific user needs.

Conclusion

The intention of this project was to design and evaluate two hybrid adaptive controllers capable of providing musical engagement and digital autonomy for individuals with limited mobility. With both prototypes comprised of various distinctive hardware and software tools, the controllers offer the potential for relatively low-cost, low latency alternatives to existing assistive technology. While undergoing the design process, I created a series of evaluative question criteria to guide the design of each prototype. Recontextualizing these in hindsight provides a clearer way to evaluate the prototypes.

The first question, involving comfort and intuitiveness, was mostly addressed through the use of a headset in each design. This is an improvement to many other accessible instruments involving movement and fatigue. Regarding being intuitive, both sip-&-puff mechanics and vocal sound triggers have been used in similar contexts, to varying success. The next question deals with range of expression, in which the first prototype was

designed to provide six triggers while the second only three. This limitation is likely the largest detriment to each design, particularly since a standard musical scale has seven notes. The subsequent two questions involve exertion reliability of inputs. Prototype I would require consistent breath control; while attainable in a small timeframe, sustained playing remains uncertain. Alternatively, the second prototype would require minimal air control, likely granting users lower exertion during use. The following question dealt with a more abstract focus, comparing a simple but intuitive design to one with a larger scope of use but likely less intuitive. For the purposes of this project, I thought it made more sense to maintain a simplistic approach, epitomized by the lack of keyboard triggers in each design. Even so, the hybrid functionalities of the prototypes certainly provide a wider range of use and expression than would otherwise be possible. The next question delves into the safety of the prototypes, where the main risk comes from the connective wiring, which, under current designs, has yet to include a safety box. By placing the microcontrollers and the connecting wiring into a small container, the user would avoid any electrical risk, while also making the device more user-friendly and aesthetic. Finally, the last criteria question involves the ease of independent setup of the controllers. Despite physical operation being straightforward, the initial software and hardware setup phase presents a clear barrier to individuals who are unaccustomed to the materials. Moreover, the audience reflects a demographic who likely have limited use of their limbs, so adversity from constructing the controller without assistance remains a clear detriment to each design.

In my interview with Dave Whalen, he identified some ideas for both designs. For Prototype I, Dave believed breath pressure to be a pivotal alternative for individuals who lack the motor control needed for physical and movement-centric controllers. He introduced the idea of mapping a more extensive series of notes to a single breath, accompanied by a visual interface for feedback. This would enable a much wider range of expression, likely at a much higher learning curve and skill ceiling. This setup would need to limit latency but could be an avenue of exploration for future projects. For

Prototype II, Dave noted that microphone setups offer a potentially less expensive, more easily adoptable alternative, but often fails when background noise is present. He offered that an in-mouth microphone or adaptive version of Peter Frampton's "Talk Box" could be more functional alternatives, particularly if adapted with an external trigger such as piano keyboard keys. When comparing the two designs, Dave seemed to perceive them as complementary rather than mutually exclusive. He noted that they might be used by both diverse and comparable audiences, even possibly combined into a single system depending on user preference and mobility needs.

Based on a consolidation of my own findings and Mr. Whalen's feedback, I have collected some recommendations for future studies and design iterations. Foremost among these is expanding the range of outputs, most basically achieved through adding another port for input. On the microphone design, adding another microphone would be simple to insert. On the breath-focused vocal design, an additional breath pressure sensor could be included with difficulty but might be more unachievable due to the nature of breath mechanics. If accomplished, however, doing so would effectively double the output capabilities. Not only would this be more useful in a musical context for playing notes but would allow the user to utilize more digital commands for common tasks. Next, to address the barrier of users needing to write their own scripts, future designs could incorporate a graphical user interface which includes customizable task commands, chord triggers, smart-home commands, or computer shortcuts. The designs would become much more user-friendly while decreasing the setup burden. An applicable future study could test user strain and fatigue over time using both designs. One of the relative unknowns with the current iterations is if they remain sustainable for long-term use, which this study would effectively address. Finally, I believe it would be useful to combine both input methods into a single controller, potentially allowing users to switch between a breath pressure sensor and a microphone depending on their specific current needs. While unfeasible for the current project, this might expand the capacity of outputs, limit user fatigue, and facilitate even greater self-autonomy.

Appendix

Sample Interview Questions

Background Info

We have a couple of adaptive controller designs we would like to show you. We want to ask some questions about the general applicability of these prototypes in the community. Of course, these are works in progress, so they are entirely flexible, and nothing is final. We are looking for feedback on how these concepts might serve a general audience, and what adjustments could make them more ergonomic.

Prototype 1

What it is: A system that uses a differential pressure sensor (sip-and-puff switch) connected to a microcontroller, which detects the direction and intensity of a user's breath.

How data gets mapped: The system allocates air pressure into six distinct triggers based on threshold and rhythm: Soft Exhale, Hard Exhale, Soft Inhale, Hard Inhale, Double Exhale, and Double Inhale. The microcontroller translates each distinct breath pattern into keyboard outputs.

What it controls: Software on the computer intercepts these keyboard keys and triggers complex actions, allowing the user to map breath patterns to musical expressions or digital commands.

Questions

I. For the general community, how useful is a breath-pressure interface as a tool to carry out repeatable actions? Why?
II. Currently, the prototype features six distinct breath patterns (soft, hard, double). For accessibility purposes, how would you describe the intuitiveness of these patterns?

III. What impact might this many triggers have on a user's fatigue over time?
IV. What are the potential benefits and drawbacks of expanding to a larger scope of use with even more triggers?
V. This design relies on software to turn simple outputs into complex actions. Based on your own experience, how practical is it to expect someone in the community to be able to customize their own setup? Why?

Prototype 2

What it is: A system with a lightweight headset microphone connected to an audio amplifier module and a microcontroller used to measure volume thresholds and sound duration.

How data gets mapped: The system would account for the volume and length of a sound while filtering out background noise. Vocal cues would be mapped to keyboard keys (ex. soft/short hum maps to F1, loud/sharp click to F2, and a long, sustained sound to F3).

What it controls: Like the first prototype, designated keys are programmed on the computer to trigger musical expressions or automated system tasks.

Questions

I. For general applicability, how useful is a vocal interface as a method for generating output? Why?
II. This prototype involves using duration and volume of sounds to map data. What unique advantages or disadvantages do you foresee when using vocal thresholds for individuals within the limited-mobility community?
III. We have talked before about microphones picking up unwanted background noise, even with a noise-filtering protocol. From your experience, what seems to be the best way to approach this issue?

IV. What specific alterations could be made to improve the ergonomics of this design?

Comparison

I. In terms of physical demands, how does exertion on the breath-pressure prototype compare to that of the vocal threshold design?
II. If the community was choosing between these two prototypes for musical expression, which one seems more universally adoptable?
III. Since people have highly diverse needs, what unique roles do you think each could fill, or in what ways do their outputs inherently address similar functionalities?

Bibliography

An open-source, breath-controlled MIDI device. (2023, May 15). *Arduino Blog*. https://blog.arduino.cc/2023/05/15/an-open-source-breath-controlled-midi-device/

Assistive technology glossary | brainline. (2017, October 30). https://www.brainline.org/article/assistive-technology-glossary

Aziz, A. D., Bursk, H., Warren, C., & Follmer, S. (2008). *The Flote: An Instrument for People with Limited Mobility*. Stanford University. https://hci.stanford.edu/publications/2008/aziz-sigaccess2008.pdf

Bhat, S. (2010). *TouchTone – An Electronic Musical Instrument for Children with Hemiplegic Cerebral Palsy*. Delft University of Technology. ResearchGate.

Blatherwick, A., & Cobb, J. (n.d.). *SenseEgg: A wireless music controller for teaching children with special learning needs*. Bournemouth University.

Campbell, J. (2020). *Skoog (Musical instrument).* EBSCO. Retrieved August 16, 2026, from https://www.ebsco.com/research-starters/music/skoog-musical-instrument

Chapin , C. M. (2020, May). *Adapting instruments, not students: A study of adaptive musical instruments*. University of Northern Iowa. https://scholarworks.uni.edu/cgi/viewcontent.cgi?article=1429&context=hpt

Chip. (2019, September 2). Synth hacker: Diy midi breath controller. *Synth Hacker*. https://synthhacker.blogspot.com/2019/09/diy-midi-breath-controller.html

Compusult Limited. (n.d.). AdaptiveSpecialties. Retrieved https://www.adaptivespecialties.com/PDF_Downloads/Jouse3%20Users%20Guide_v3.0.pdf

Davanzo, N., Dondi, P., Mosconi, M., & Porta, M. (2018). Playing music with the eyes through an isomorphic interface. *Proceedings of the Workshop on Communication by Gaze Interaction*, 1–5. https://doi.org/10.1145/3206343.3206350

Davanzo, N., Filippis, M. D., & Avanzini, F. (2021). *Netychords: An Accessible Digital Musical Instrument for Playing Chords using Gaze and Head Movements*. University of Milan. ScitePress. https://www.scitepress.org/Papers/2021/106892/106892.pdf

Dixon, K. (2003). *Physical Tools Needed for Playing a Musical Instrument*. Kodaly Music Education Institute of Australia. Informit. https://search.informit.org/doi/epdf/10.3316/informit.092896752728848

Duarte, E. G., Cossette, I., & Wanderley, M. M. (2023). Analysis of Accessible Digital Musical Instruments through the lens of disability models: A case study with instruments targeting d/Deaf people. *Frontiers in Computer Science*, *5*. https://doi.org/10.3389/fcomp.2023.1158476

Faschi, V. (2025). *UniMIDIHub: A Prototype of Accessible Digital Musical Instruments for Musical Inclusion*. University of Milan. https://ceur-ws.org/Vol-4151/short13.pdf

Finch, M., Quinn, S. L., & Waterman, E. (2016). Improvisation, adaptability, and collaboration: Using aumi in community music therapy. *Voices: A World Forum for Music Therapy*, *16*(3). https://doi.org/10.15845/voices.v16i3.834

Frid, E. (2019, July 26). *Accessible digital musical instruments—A review of musical interfaces in inclusive music practice*. KTH Royal Institute of Technology. MDPI. https://www.mdpi.com/2414-4088/3/3/57

Graham-Knight, K., & Tzanetakis, G. (2015). Adaptive music technology using the Kinect. *Proceedings of the 8th ACM International Conference on PErvasive Technologies Related to Assistive Environments*, 1–4. https://doi.org/10.1145/2769493.2769583

Grond, F., Shikako-Thomas, K., & Lewis, E. (2020). *Adaptive musical instruments (Amis): Past, present, and future research directions*. https://doi.org/10.15353/cjds.v9i1.598

Holland Bloorview's Hummingbird assistive technology – a Canadian first – gives kids, adults with disabilities a voice. (2022, September 21). Holland Bloorview. https://hollandbloorview.ca/stories-news-events/news/holland-bloorviews-hummingbird-assistive-technology-canadian-first-gives

Instructions | usb midi breath controller | hackaday. Io. (n.d.). Retrieved August 16, 2026, from https://hackaday.io/project/161678/instructions

Integrated silicon pressure sensor on-chip signal conditioned, temperature compensated and calibrated. (n.d.). STMicroelectronics. Retrieved https://www.st.com/en/mems-and-sensors/mpx5010dp.html

Isatma. (n.d.). Retrieved August 15, 2026, from https://sites.google.com/view/isatma

Jamboxx. (n.d.). Retrieved August 15, 2026, from https://sites.google.com/view/jamboxx/home

Lamont, A. (2012). Emotion, engagement and meaning in strong experiences of music performance. *Psychology of Music*, *40*(5), 574–594. https://doi.org/10.1177/0305735612448510

Makandar, A., & Halalli, B. (2015). *Image Enhancement Techniques using Highpass and Lowpass Filters*. International Journal of Computer Applications. ResearchGate. https://doi.org/10.5120/19256-0999

Manzo, V. J. (n.d.). *The ratlab*. The RATLab Is a Research Collective at WPI Focusing on the Convergence of Music and Technology through the Electric Guitar Innovation Lab (EGIL) and the Interactive Music Systems Lab (IMSLab). Retrieved August 15, 2026, from https://theratlab.org/

MPS20n0040d ported pressure sensor breakout board. (n.d.). Retrieved August 16, 2026, from https://www.twinschip.com/MPS20N0040D_Breakout_Board

Quadstick User Guide. (n.d.). Retrieved August 16, 2026, from https://quadstick.s3.amazonaws.com/documents/user_manual/um/introduction.htm

Sip/puff switch. (n.d.). Retrieved August 15, 2026, from https://www.orin.com/access/sip_puff/

The Magic Flute II. (n.d.). Retrieved https://mybreathmymusic.com/wp-content/uploads/2022/03/Magic-Flute-manual-eng-2018.pdf

Tzanetakis, G., & Graham-Knight, K. (2015, July 1). *Adaptive music technology using the Kinect*. PETRA '15: Proceedings of the 8th ACM International Conference on PErvasive Technologies Related to Assistive Environments. ACM Digital Library. https://dl.acm.org/doi/epdf/10.1145/3206343.3206350

Using Switches to Access Button Presses | Video. (n.d.). GameAccess. Retrieved https://gameaccess.info/using-switches-to-access-button-presses/

Ward, A., Davis, T., & Bevan, A. (2019, July 18). *Music Technology and Alternate Controllers for Clients with Complex Needs* . Music Therapy Perspectives. Oxford Academic. https://doi.org/https://doi.org/10.1093/mtp/miz006

Wobbrock, J. O., Gajos, K. Z., Kane, S. K., & Vanderheiden, G. C. (2018). Ability-based design. *Communications of the ACM*, *61*(6), 62–71. https://doi.org/10.1145/3148051

Zatorre, R. J. (2018). Why do we love music? *Cerebrum: The Dana Forum on Brain Science*, *2018*, cer-16-18. https://pmc.ncbi.nlm.nih.gov/articles/PMC6353111/

Zhang, Y., Li, Y., Chin, D., & Xia, G. G. (n.d.). *Adaptive Multimodal Music Learning via Interactive-haptic Instrument*. New York University. Arxiv. Retrieved https://arxiv.org/pdf/1906.01197